



Society of Petroleum Engineers

SPE-227831-MS

AI-Driven Petrophysical Predictions: Optimizing Performance in Reduced Dataset Environments

Ramsin Y. Eyvazzadeh and N. Almulhim, Saudi Aramco

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227831-MS](https://doi.org/10.2118/227831-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The objective of the study is to enhance the prediction of permeability by leveraging of advanced artificial intelligence technologies in reservoirs with limited datasets. Initially, the paper will focus on employing transfer learning techniques to establish analogies between the current reservoir and geologically similar reservoirs with larger datasets. In cases where no suitable analogy could be identified, ensemble methods were utilized to enrich the dataset by constructing robust predictive features. To address imbalanced datasets, the study focused on using data augmentation to expand the current dataset. These approaches aim to optimize performance, reduce uncertainty and deliver robust predictions in data constraint reservoirs.

The study presented in this paper uses a multi-faceted approach to address the challenges of petrophysical attributes prediction in reservoirs with limited datasets. First, several transfer-learning tools, including Procrustes disparity, Kullback-Leibler divergence, Silhouette scores analyses are applied robustly identify analogies between the target reservoir and external datasets. For the reservoirs where no analogies exist, ensemble methods are used to generate robust synthetic features to enrich the dataset. In datasets that may be biased or imbalanced, techniques such as Synthetic Minority Oversampling Technique (SMOTE) are employed to create representative samples. This workflow integrates these techniques into a streamline process, starting with data processing, analysis, followed by model training and validation using statistical metrics to ensure reliability and robustness of the predictions.

The deployment of these AI methodologies in permeability prediction has yielded promising results. The application of transfer learning techniques has clearly identified analogies between reservoirs, significantly improving prediction accuracy. In reservoirs lacking direct analogies, ensemble and data augmentation methods have effectively reduced bias and enhanced model robustness. A study conducted on number of fields have shown that models trained using this methodology have greatly improved applicability and performance compared to traditional methods.

This study introduces a novel framework that moves beyond traditional approaches to identify analogies in reservoirs, which often relied on statistical comparisons or subjective geological expertise. By integrating these AI methods, the study provides a quantifiable and objective method to establish a deeper understanding of relationships between reservoirs, allowing for improved predictions. The study also addresses the critical point where data scarcity affects model reliability and how data augmentation and ensemble methods can

be employed to improve model performance. The ability to integrate these advanced tools represents a significant leap forward in decision-making processes in field development and management.

Introduction

The integration of machine learning algorithms and automation in petrophysical analysis has significantly improved traditional workflows, particularly in properties prediction, where core, well log, and dynamic data are utilized to develop 3D static and simulation models. Nevertheless, many practical petrophysical applications are hindered by the limited availability of labeled data from core-measured properties, resulting in reduced dataset environments. This constraint poses a challenge to the reliability and performance of standardized AI algorithms, which typically require large datasets to be effective. As a consequence, models derived from these limited datasets can introduce substantial uncertainties, leading to decisions to either acquire additional data or optimize the existing dataset to mitigate these uncertainties.

This paper focuses on the development and optimization of AI-driven petrophysical predictions in environments, where data is incomplete, scarce, or lacks comprehensive representation of an entire field, particularly in heterogeneous and complex reservoirs.

In order to address these challenges, we initially introduce diagnostic tools – including learning curves and cross validation performance techniques to identify models that maybe constraint by limited labeled data. We then illustrate three strategies that improve performance: (1) **Transfer learning** which identifies datasets from nearby fields or similar geological settings to supplement the existing dataset; (2) **ensemble models**, which improve model diversity and reduce variance and improve model robustness; and (3) **data augmentation**, which synthetically expands and enriches the dataset to better capture the feature space.

Overfitting, Bias, and Misleading Statistical Indicators in Reduced Datasets

One of the biggest challenges posed in permeability modeling is the tradeoff between bias and variance, and this is especially critical in reduced datasets. Models with high bias tend to be very simplistic and often fail to capture the non-linear relationships inherent between petrophysical properties, well log data, and geological depositional and diagenetic facies. Figure 1 shows an example of a model where the data is split between two different training and test sets. In a high-bias model, the training and test errors between predicted and measured values do not change much when using different training datasets.

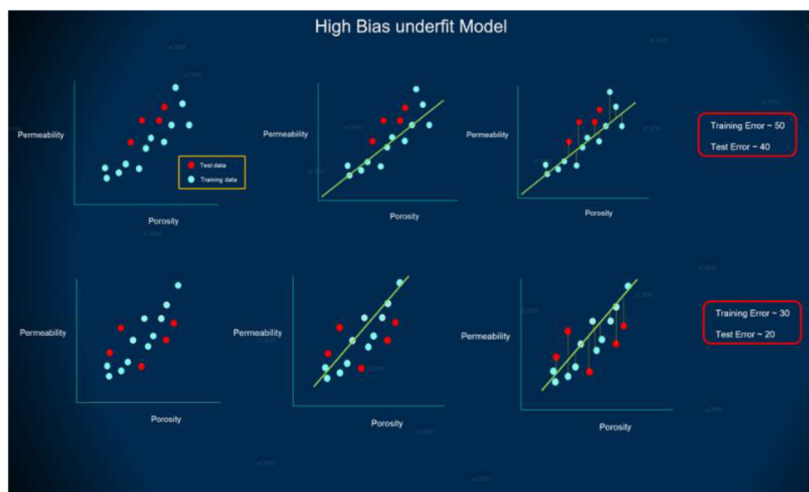


Figure 1—A high-bias model where the training and test errors are compared using a different split between training datasets. Note the small changes between the two models.

Conversely, in models that exhibit high variance (Figure 2), the errors associated with the training data are generally low, nearly zero (overfitting), and the test errors between different datasets have a high degree

of variability, with significant changes as different splits are chosen between training and testing datasets. These models are often too complex relative to the data size, capturing noise and spurious patterns that do not generalize well.

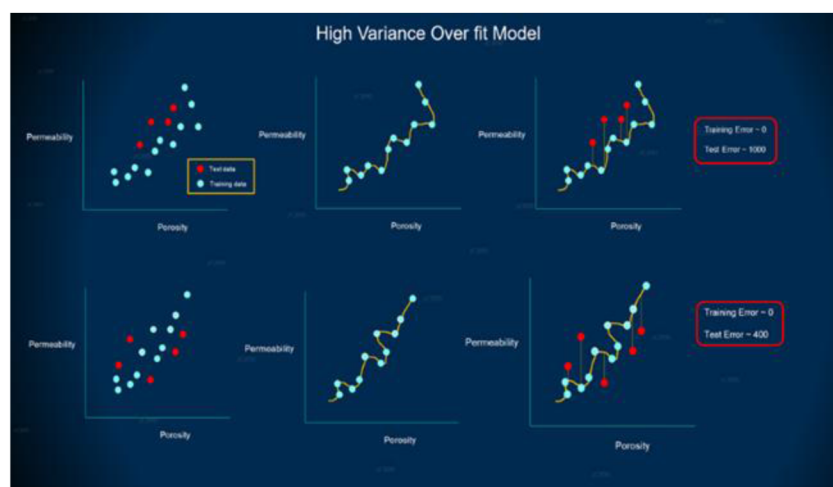


Figure 2—A High variance model where the training and test errors are compared using a different split between training datasets. Note the large difference in training errors by using different datasets.

In reduced dataset environments, common performance metrics such as R^2 , RMSE and p-values can be misleading in providing model accuracy. With sparse core data, R^2 may appear to be high and provide a predictive strength of the model, while its only showing the overfit. Similarly, RMSE can be skewed to outliers, making it unreliable in small samples without confidence intervals or repeated validation. In regression models, p-values can often be misinterpreted as indicators of feature relevance and can greatly fluctuate due to multicollinearity and limited degrees of freedom. Even feature rankings in ensemble models can be dominated by correlated logs, further complicating interpretation.

Identifying reduced (Sparse) datasets

Although various conventional methods are available to assess data adequacy, such as coverage maps that evaluate the representation of labeled data across different depositional and diagenetic geological facies, stratigraphic units, and depth intervals, these tools offer only a static and superficial view of data sufficiency. Additional techniques, including permutation importance and stratified residual analysis, can provide further insights by revealing whether predictive values hold consistent across different data subsets and identifying biases in individual facies due to imbalanced sampling.

However, these methods have limitations, as they may overlook issues related to a model's ability to learn and generalize. A more effective approach is to utilize statistical and diagnostic models which provide model-specific insights into whether the dataset contains sufficient representative information to develop robust and predictable models. These include bias-variance trade-off analysis, tracking of validation error metrics, statistical sampling checks, model uncertainty estimation and sample size sensitivity analysis. In addition to these methods, two of the most powerful methods in reduced dataset environments are learning curves and cross validation performance evaluation. By leveraging these tools, it is possible to gain a more comprehensive understanding of the data's strengths and weaknesses, ultimately leading to more reliable and accurate model development.

Learning Curves and Cross Validation Performance

In small datasets, where the number of core permeability samples may be limited, learning curves and cross validation performance metrics are some of the most effective tools for assessing model behavior and

determining whether the available data is sufficient to accurately capture the complexity of the modeled reservoir. Figures 3 and 4 illustrate a series of learning curves generated using K-fold cross-validation, where different splits of training and testing data were applied to both a large and a small dataset. Typically, the plots are generated where the number of samples is plotted versus an error metrics such as RMSE, MAE or R^2 , providing a visual representation of how model performance changes as the dataset size increases.

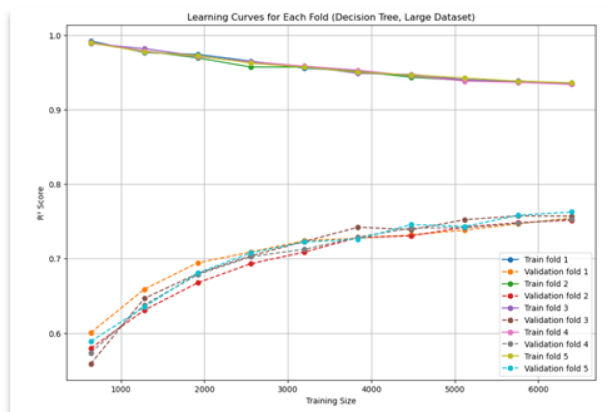


Figure 3—Learning Curves with cross validation on a large dataset.

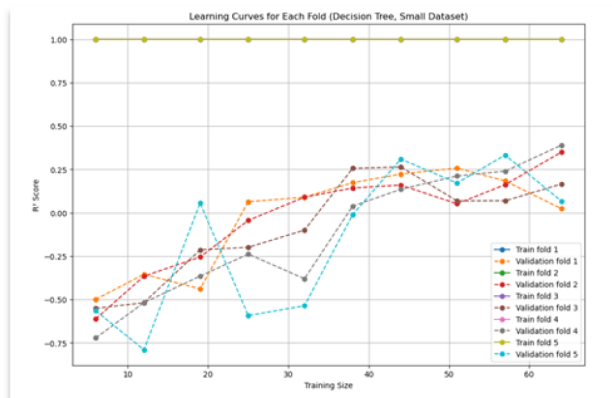


Figure 4—Learning Curves with cross validation on a small dataset.

These curves reveal some critical insights:

- If the training error is low validation error remains high, the model is likely overfitting (Figure 4)
- If training and validation errors are high, this suggests underfitting, often due to a simplistic high bias model
- A persistent gap between training and validation error suggest insufficient data diversity, rather than just the quantity
- If the models perform with spurious validation errors, this may suggest that this variability may be highly dependent on specific subset of the data that it was trained on.
- If the learning curves plateau cross validation is stable but poor, the model may have irreducible error and the issue may not be data quantity, but may require more feature engineering.

Transfer Learning

If it has been determined that the present dataset is insufficient to support a robust model, the most direct way is to perform a gap analysis and acquire additional measurements. However, in many real-world

scenarios, this may not be feasible, due to high operational cost or lack of drilling opportunities. In such cases a practical solution is to leverage data from analog fields with similar petrophysical, depositional and diagenetic characteristics with the target reservoir.

Transfer learning algorithms play an important role in quantitatively assessing the compatibility between datasets. When log data is available from both source and target fields and core data is not available in the field of the study, a range of statistical and geometric fields can be used to compare their feature spaces using well log data. Procrustes disparity, Kullback-Leibler (KL) divergence, and Silhouette scores are three techniques that have been found to be particularly applicable in petrophysical applications, offering unique advantages in evaluating the similarity and differences between feature spaces. While other techniques, such as Mahalanbois distance and Eigenvector similarity, are also available, the first three methods are well-suited for petrophysical data analysis due to their ability to handle complex and high-dimensional data. These techniques ensure that the statistical and physical behavior of the two datasets are sufficiently aligned.

If a high degree of similarity is established with all of these methodologies, labeled data from the analog field can be used to develop models in the target field. This strategy enhances model viability in a data-sparse fields without the need for costly acquisition programs.

Procrustes Disparity

The analysis is a geometric method that assess the similarity between two sets of multivariant data by an optimal alignment. First emerging in psychometrics and shape analysis, the method has found emergence in machine learning and geoscience applications. In petrophysics, it offers a systematic way of comparing petrophysical properties between two fields. It has been very effective where core data may not be available in one field and well log data can be used to make this assessment. The following images (Figure 4) illustrate the workflow steps:

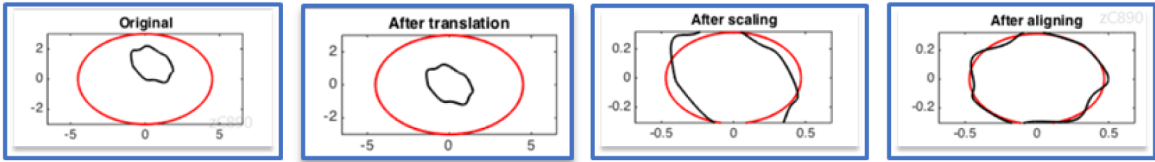


Figure 5—A step-by-step workflow of calculating residuals using Procrustes disparity.

- Start with the same number of columns between two datasets
- Each dataset is centered to the mean, so the average position is the same in space(translation).
- Scale the data to a unit norm, where the variance is standardized.
- The source data is rotated to the target data dataset.
- Measure the residual difference and calculate the disparity

The following table lists heuristic guidelines that should be supplemented geological reasoning and other statistical comparisons.

Table 1—A heuristic guide for thresholds for PD analysis.

Procrustes Disparity	Interpretation	Transferebility Confidence
0.00 - 0.05	Very strong similarity	Excellent
0.05 - 0.15	Moderate similarity	Likely transfereable
0.15 - 0.30	Noticeable differences, verify with other methods or geology	Caution advised
>0.30	Highly dissimilar	Not recommended

Important considerations when using this analysis:

- The number of dimensions and the number of samples can affect the number disparity. Very high-dimensional spaces may give misleading low or high disparity (curse of dimensionality) unless the data is preprocessed properly.
- Consider facies driven stratified models in very heterogenous environments.
- If the disparity is borderline, use additional measures explained in the following sections

Kullback-Leibler Divergence: Distributional difference

A fundamental tool that quantifies how one distribution diverges from a second or reference distribution. Described as measure of information loss, it describes the inefficacy of approximating a true distribution with an alternate distribution for every feature in the database.

The following are histogram differences between two datasets that include, porosity, volume of calcite, sonic travel time and formation density.

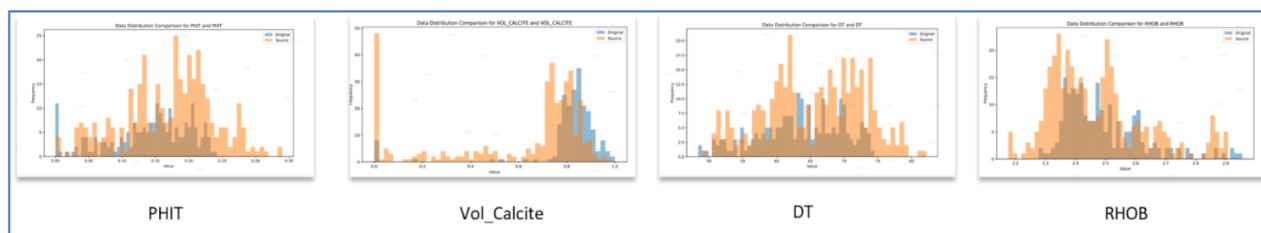


Figure 6—A step-by-step workflow of calculating mismatch using KL divergence. (Source dataset: Orange, Target dataset: Blue, Overlap: brown).

The following outline steps in the workflow:

- Feature selection
- Estimate the distributions in both datasets using histograms with consistent binning and Kernel Density Estimation (KDE).
- Normalize the distributions to ensure the sum is equals to 1.
- Apply the KL divergence across the bins or intervals
- Interpret the results.

There are no universal thresholds, but the following should be as a guideline.

Table 2—A general guideline for KL divergence.

KL Divergence	Interpretation	Transferebility Confidence
<0.1	Very strong similarity	Excellent
0.1 - 0.3	Moderate similarity	Likely transfereable
0.3-1.0	Noticeable differences, verify with other methods or geology	Caution advised
> 1.0	Highly dissimilar	Not recommended

Important considerations when using this analysis:

- Choice of probability estimation can skew KL values. Use the same for both datasets (KDE or histogram)

- In high dimensional data use dimensionality reduction techniques to reduce redundant divergence calculations.
- Imbalance in datasets, where one dataset contains more data than the other may bias the data, down sample the data or perform the analysis using geological facies classifications
- KL divergence calculations are highly sensitive to outliers.
- Comparing distributions without considering depositional and/or diagenetic sequences can be misleading.
- KL divergence is asymmetric, use Jensen-Shannon divergence (JSD) for symmetric and smooth features.

Silhouette Score: Cluster Quality and Separation

The Silhouette score is widely used metric that quantified how well each data point fits within a cluster compared to other clusters. It evaluates cohesion (how close a point is to other in the cluster) and separation (how far a point is from points from other clusters). The analysis is particularly useful in petrophysical studies assessing the quality of log-based clustering.

The data is originally clustered using DBSCAN, K-means or hierarchical clusters into k-clusters. A silhouette score is calculated by mean distances of the point to the inner clusters and all of the points in neighboring clusters. The overall score is then calculated by the mean of the individual scores.

Table 3—A general guideline for Silhouette Score.

Silhouette Score	Interpretation	Transfereability Confidence
0.70-1.00	Strong structure: well defined separated clusters	Excellent
0.50-0.70	reasonable structure, moderate separation	Likely transfereable
0.25-0.50	Weak Structure, clusters overlap	Caution advised
< 0.25	No substantial structure, clustering maybe random	Not recommended

Important considerations:

- As the score depends on distances, all variables should be normalized before clustering.
- The model works best with convex clusters, rather than spherical. Clustering methods such as DBSCAN and hierarchical clustering are best suited for these datasets.
- Outliers can significantly lower Silhouette scores. Preprocessing with outlier removal is highly recommended.
- The score can artificially improve or degrade depending on the chosen number of clusters. Plotting the score versus number of clusters is a common method to determine the number of clusters.
- Very effective in geological facies validation, using discrete inputs.

Ensemble Methods

When transfer learning proves ineffective in identifying analogies, ensemble methods offer a powerful alternative for improving model performance in reduced dataset environments. Unlike transfer learning, which relies on outside data sources, ensemble techniques work by combining multiple models to extract more robust predictions from the available data. Methods like Bagging, Boosting and stacking help to reduce variance, improve bias, mitigate overfitting and improve performance, particularly when single models struggle by labeled sample sizes.

Bagging

Bagging focuses on reducing variance by training multiple models in parallel on different subsets of the training data. Each subset is generated by bootstrapping technique, where samples are chosen and replaced, where data subsets can have the same samples. The classic example of this method is Random Forest, which builds multiple trees based on different bootstrapped datasets and averages the output for regression or majority vote for classification. On average, 63% of the data is used in each bootstrap sample. The remaining ~37% Out-of-Bag (OOB) data serves as a validation set, that is used as validation to evaluate model performance. By aggregating OOB predictions, model performance can be unbiasedly evaluated by estimating the final model prediction error. Figure 7 is a illustration of the bagging techniques, where multiple bootstrap datasets were generated to develop models that were evaluated and validated using the OOB samples.

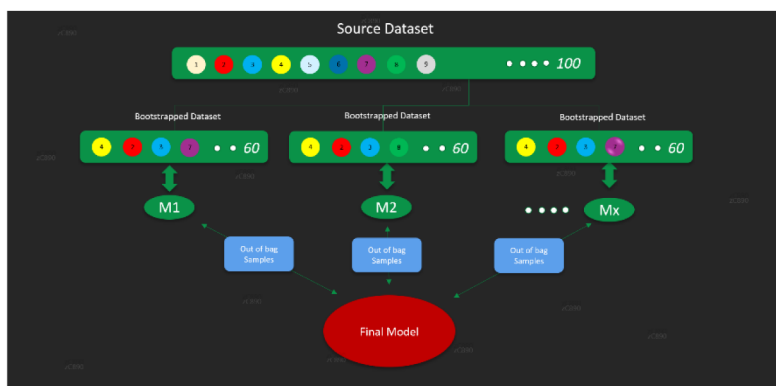


Figure 7—Bagging, where multiple models are developed based on bootstrap datasets and validated using OOB samples.

The strength of bagging is the ability to reduce variance, especially in small noisy datasets. By training diverse subsets, the model's ability to recognize different patterns is increased and the aggregation step reduces issues with overfitting. Bagging is simple, scalable and relatively robust to outliers, making it effective in petrophysics, but maybe limited in very complex morphologies.

Boosting

While bagging builds models independently, boosting builds them sequentially. Each model is developed to reduce errors from the previous model with a focus on hard to predict samples. It makes this model more powerful in reducing bias and improving accuracy in complex fields, but more susceptible to overfitting. The model required careful regularization.

Stacking

Stacking takes a different approach by combining multiple diverse base models – often of different types, such as linear, decision trees and neural nets and combines them into a meta model to learn how to combine the predictions. By combining the complementary strengths of different models, it allows to exploit the best patterns recognized by each model. Even though the methods are more complex and data intensive, they often outperform bagging and boosting models, especially in capturing complex geological and petrophysical behavior.

Figure 8 presents a comparative analysis of two machine learning approaches for predicting core-measured permeabilities from well log data: A K-Nearest Neighbors (KNN) model and a stacking model. The stacking model, which employs a meta-learning strategy to combine the predictions of multiple base models, showed significant improvement over the KNN model in terms of accuracy and robustness. This is attributed to the stacking model's ability to capture complex patterns and relationships in the data, as well as its capacity to mitigate the effects of overfitting and underfitting. The use of multiple base models in the

stacking ensemble allows for a more comprehensive representation of the data, resulting in a more accurate and reliable prediction of permeabilities. The following cross-plots illustrate the results of a blind test, where the predicted permeabilities are compared to the core-measured values, demonstrating the superior performance of the stacking model.

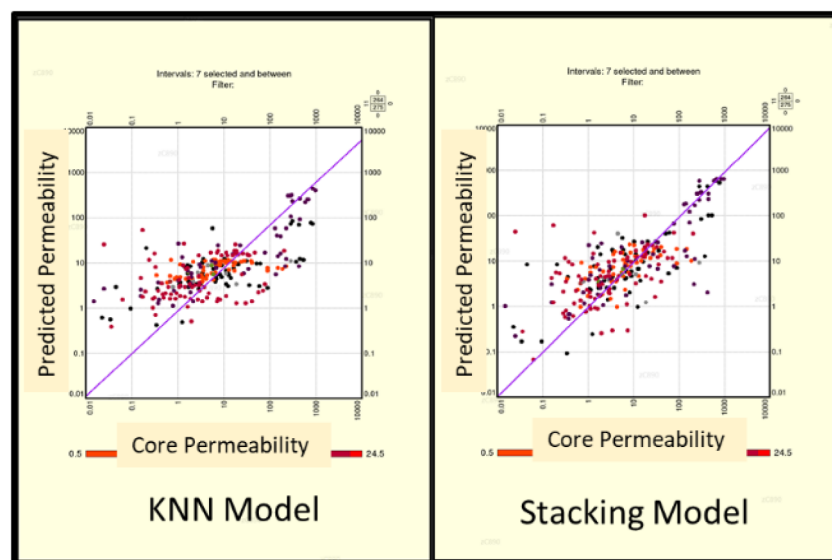


Figure 8—A comparison between KNN and Stacking core measured permeability with well log data.

Data Augmentation

Another approach in improving performance in reduced and especially imbalanced datasets is to augment the data, where certain classes or geological facies are not well presented. Data augmentation involves synthetically generating samples to increase volume or diversity without the need of finding analogies or acquiring more physical measurements. In petrophysical modeling, where core measurements are used as labeled data, this method is especially useful. Techniques such as SMOTE (Synthetic Minority Over-sampling Technique), ADASYN (Adaptive Synthetic Sampling) or Gaussian noise Injection can be applied to derive synthetic data in the minority classes, helping the model with more balance and representation and less bias.

SMOTE augments the data by linear interpolation between each minority observation and its k-nearest neighbors. Great care should be taken when using these methods, especially when the relationship between the labeled data (typically, core data) and the feature datasets are not linear, especially in complex heterogenous formations. ADASYN improves on SMOTE by targeting the areas where the model performs poorly, generating samples in underrepresented zones. This targeted approach reduces the risk of overfitting, but just like SMOTE, the method is sensitive to noise in the original dataset. Gaussian noise injection perturbs existing data points by adding small, random variations drawn from a normal distribution. When working with very small datasets, a hybrid approach by combining Gaussian injection, SMOTE and ADASYN, guided by geological constraints, can offer more a more robust augmentation strategy.

Summary

Accurate petrophysical predictions - such as estimating permeability models in a 3D static models, are frequently limited by scarcity of core data. In many cases, machine learning models are developed in insufficient labeled data, resulting high uncertainty and reduce confidence in predictions. This paper presents a well-structured AI driven workflow to optimize and enhance performance in reduced dataset environments. Initially, it discussed the bias-variance tradeoff and how data scarcity impacts generalization.

Learning curves and cross validation techniques are then introduced to identify when a dataset is too small to support a robust model. In situation where, additional data is available from nearby fields, transfer learning techniques are introduced to quantitatively identify analogies. When such analogs cannot be identified, the paper explores ensemble techniques to exploit more out of the existing dataset and reduce model variance. In extreme cases, where neither analog transfer or ensemble approaches are viable, data augmentation techniques are proposed to expand the training dataset. Collectively, these strategies offer practical solutions for delivering AI-driven petrophysical predictions in data-constrained environments.

References

1. Kohavi, R. 1995. A Study of Cross-Validation and Bootstrap for Accuracy Estimation and Model Selection. *Proc., 14th International Joint Conference on Artificial Intelligence*, 1137-1143.
2. Geman, S., Bienenstock, E., and Doursat, R. 1992. Neural Networks and the Bias/Variance Dilemma. *Proc., 1992 IEEE International Conference on Neural Networks*, 1-8.
3. Amari, S. 1993. Learning Curves and Optimal Learning Rates. *Neural Networks*, **6** (5): 661-674. doi: [10.1016/S0893-6080\(09\)80004-2](https://doi.org/10.1016/S0893-6080(09)80004-2)
4. Freund, Y., and Schapire, R. E. 1997. A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting. *Journal of Computer and System Sciences* **55** (1): 119-139. <https://doi.org/10.1006/jcss.1997.1504>
5. Tibshirani, R. 1996. Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society: Series B* **58** (1): 267-288. <https://doi.org/10.1111/j.2517-6161.1996.tb02080.x>
6. Chawla, N. V., Bowyer, K. W., Hall, L. O., and Kegelmeyer, W. P. 2002. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research* **16**: 321-357. <https://doi.org/10.1613/jair.953>
7. He, H., Bai, Y., Garcia, E. A., and Li, S. 2008. ADASYN: Adaptive Synthetic Sampling Approach for Imbalanced Learning. In *Proceedings of the IEEE International Joint Conference on Neural Networks*, 1322-1328. <https://doi.org/10.1109/IJCNN.2008.4633969>
8. Pan, S. J., and Yang, Q. 2010. A Survey on Transfer Learning. *IEEE Transactions on Knowledge and Data Engineering* **22** (10): 1345-1359. <https://doi.org/10.1109/TKDE.2009.191>
9. Cover, T. M., and Thomas, J. A. 2006. Elements of Information Theory, 2nd ed. Wiley-Interscience, New York.
10. Al-Saedi, A. G., Rezaee, R., and Kadkhodaie, A. 2021. A Machine Learning Workflow for Estimating Permeability in Tight Carbonate Reservoirs Using Core and Log Data. *SPE Reservoir Evaluation & Engineering* **24** (2): 372-384. SPE-200246-PA. <https://doi.org/10.2118/200246-PA>